

Accepted Manuscript

Title: 3D nitrogen-doped graphene aerogel: A low-cost, facile prepared direct electrode for H₂O₂ sensing

Author: Zhi-Xiong Cai Xin-Hong Song Yi-Ying Chen Yi-Ru Wang Xi Chen



PII: S0925-4005(15)30272-0
DOI: <http://dx.doi.org/doi:10.1016/j.snb.2015.08.094>
Reference: SNB 18949

To appear in: *Sensors and Actuators B*

Received date: 16-6-2015
Revised date: 11-8-2015
Accepted date: 21-8-2015

Please cite this article as: Z.-X. Cai, X.-H. Song, Y.-Y. Chen, Y.-R. Wang, X. Chen, 3D nitrogen-doped graphene aerogel: a low-cost, facile prepared direct electrode for H₂O₂ sensing, *Sensors and Actuators B: Chemical* (2015), <http://dx.doi.org/10.1016/j.snb.2015.08.094>

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

3D nitrogen-doped graphene aerogel: a low-cost, facile prepared direct electrode for H₂O₂ sensing

Zhi-Xiong Cai^{1†}, Xin-Hong Song^{1†}, Yi-Ying Chen¹, Yi-Ru Wang^{1*}, Xi Chen^{1,2*}

¹ MOE Key Laboratory of Spectrochemical Analysis & Instrumentation, College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, 361005, China.

² State Key Laboratory of Marine Environmental Science, Xiamen University, Xiamen, 361005, China.

Abstract

A facile, template-free and low-cost strategy has been developed to construct a 3D nitrogen (N) doped graphene aerogel (GA) by the incorporation of dopamine (DA). 3D-NGA can be served as a highly efficient electrocatalyst in the reduction of H₂O₂, which is then applied in the nonenzymatically electrochemical determination of H₂O₂. The structure and morphology of hollow 3D-NGA was characterized by transmission electron microscopy (TEM), scanning electron microscopy (SEM) and X-ray photoelectron spectroscopy (XPS). Electrocatalytic activity of 3D-NGA was investigated by cyclic voltammetry (CV) and chronoamperometry, exhibiting a direct for the reduction of H₂O₂ at -0.5 V. This constructing sensor was applied in detection of H₂O₂ with a detection limit of 0.05 mM, a linear detection range from 0.2 to 35 mM. Due to superior catalytic performance and low cost of 3D-NGA electrode, it shows great potential in applications for direct sensing of H₂O₂.

Keywords: 3D N-doped graphene aerogel; Electrocatalyst; Sensing of H₂O₂; Oxygen Reduction Reaction.

1. Introduction These authors contributed to the work equally

Carbon-based materials, especially carbon nanotubes and graphene have attracted tremendous interest in a variety of fields due to its unique structure and electronic properties [1, 2]. Most

[†] These authors contributed to the work equally.

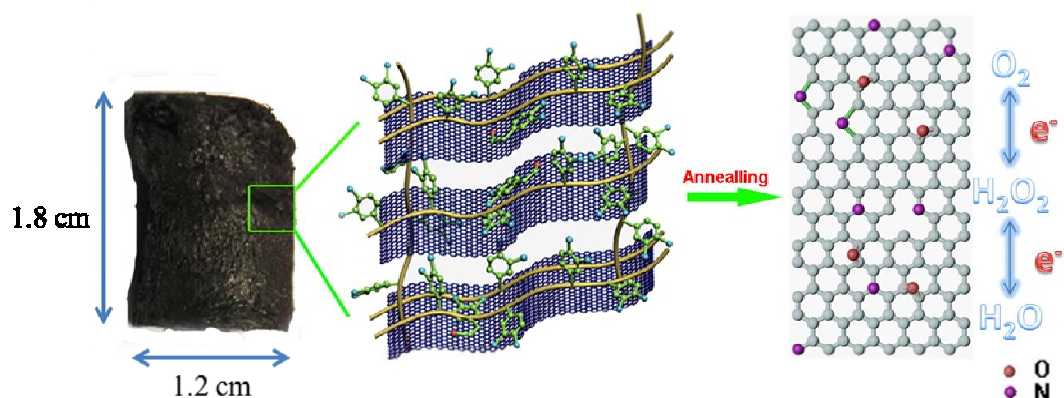
* Corresponding author. Tel: +86 592 2184530; Fax: +86 592 2184530;
E-mail address: xichen@xmu.edu.cn

notably, in the area of electrochemical analysis, carbon-based sensors show good performance in the detection of molecules such as uric acid [3], ascorbic acid [4], dopamine [5], nitric oxide [6], H_2O_2 [7], H_2S [8], glucose [9] and TNT [10]. Recently, the researches of graphene-based nanocomposites on biosensors have been greatly developed. However, these nanocomposites meet some troubles due to small specific surface area and poor durability, which are ascribed to the inevitable aggregation between individual graphene sheets driven by strong π - π stacking interaction [11]. In particular, aerogels, constructed by the integration of the individual graphene sheets as nanoscale building blocks into macroscopic structure, have shown a thriving development due to their excellent properties. Compared with the conventional graphene sheets, such 3D graphene aerogels offer large active surface area, high conductivity, unhindered substance diffusion, and the multiple electron paths, which provide great potential in applications, such as biosensor [5].

Chemical doping with foreign atoms, such as nitrogen atoms, a intrinsic modification of graphene, is an effective strategy to modulate electronic properties and surface chemistry of assembled graphene by the regulation of the carbon-carbon bonds. The doping improves its physicochemical characteristics, especially its electrochemical performance [12]. Oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) are fatal in the rechargeable metal-air batteries and regenerative fuel cells [13]. Noble metals [14] and transition-metal oxides [15] are commonly utilized as the catalysts. However, high cost, scarcity, and inferior durability of noble metals, and complicated fabrication of transitionmetal oxides with low conductivity largely limit their practical applications. In contrast, non-metal materials consisting of earth-abundant elements (C, H, O, and N) that act as catalytically active components have been recently recognized as a new class of low-cost and environmentally benign electrocatalysts [16-18]. Qu et al. first reported that N-doped graphene act as a metal-free catalyst possesses much better electrocatalytic activity, long-term stability, and tolerance to crossover effect than platinum towards ORR [19]. Zhao et al. reported that the N-doped carbon materials function as the efficient OER electrocatalysts, which are comparable to those of iridium and cobalt oxide catalysts and are the best among the non-metal OER electrocatalysts [16]. It has been also reported that N-doped graphene can promote the electrochemical reduction of H_2O_2 and direct electron transfer kinetics of glucose oxidase [20]. It has also been hypothesized that a synergistic support effect is with

N-graphene because the nitrogen doped carbon nanostructures are known to decompose reactive intermediates such as hydrogen peroxide into oxygen formed during ORR [21]. Generally, there is an intrinsic correlation between the redox of H_2O_2 and ORR, and the electrocatalyst with a low overpotential for ORR should be favorable for H_2O_2 redox [22]. This suggests a possibility to greatly improve the performance of the biosensor for the detection of H_2O_2 by exploring suitable N-doped graphene. Briefly, the N-doped graphene would show high sensitivity in H_2O_2 biosensing applications and excellent electrocatalytic activity for ORR and OER.

In addition, the determination of H_2O_2 has received considerable attention due to its importance in the fields of food, pharmaceutical, clinical, industrial and environment protection [23, 24]. Among various analytical techniques, electrochemical H_2O_2 sensors with electrocatalysts hold a leading position among various biosensors owing to their high sensitivity, good selectivity and easy handling [24]. Although noble metal combining graphene hybrids with synergy or multiple functionalities act as mimic peroxidase, and exhibit excellent sensitivity and selectivity towards H_2O_2 , it still suffers from some drawbacks as high cost and low poison resistance. In this work, based on our previous work [5], a facile, template-free and low-cost strategy was used to design a 3D nitrogen (N) doped graphene aerogel (GA) by incorporating chemical motif of dopamine (DA) (scheme 1). Combining the 3D architecture, exceptional properties of graphene and surface mediation, the metal-free 3D NGA exhibited good electrocatalytic activity towards H_2O_2 reduction. Furthermore, a novel amperometric sensor for H_2O_2 was constructed by using the metal-free 3D NGA as direct electrode, taking advantage of N-doped nature as catalysts and its chemical stability. Results revealed that 3D NGA the exhibit a remarkable electrocatalytic activity for the reduction of H_2O_2 without loading any enzymes. The electrocatalytic performance of 3D-NGA towards O_2 reduction/evolution reactions is also discussed.



Scheme 1. Schematic illustration of the preparation of and electrocatalysis process of the 3D-NGA.

2. Experimental

2.1. Materials

The graphite powder was purchased from Lvyin Co. (China). 5% Nafion ethanol solution and dopamine (DA) were from Aldrich Chem. Co. (USA). All other reagents were purchased from Sinopharm Chemical Reagent Ltd. Co. (China), and were of analytical grade and used without further purification. The pure water from a Millipore Autopure WR600A system (USA) was used for the solution preparation.

2.2. Preparation

Graphene oxide (GO) sheets were synthesized based on the well-known Hummers' method but with some modification [25]. The 3D-NGA hydrogel was prepared using the following procedure: typically, 15 mg DA was added into a GO aqueous dispersion (15 mL, 1 mg mL⁻¹). The dispersion pH was adjusted to about 8.0, and then sonicated for 40 min. The mixture was sealed in a 20 mL Teflon-lined autoclave and maintained at 180 °C for 12 h. After cooled to room temperature, the as-prepared hydrogel was washed with ethanol and water and then freeze-dried. Ultralight NGA was obtained after heating the freeze-dried graphene gel at 800 °C for 3 h under an Ar atmosphere. Simultaneously, 3D GA was synthesized by the same method in the absence of DA. To construct an electrode, the freestanding of cylindrical 3D NGA was cut into piece (1 cm × 0.5 cm, 0.3 mm thick). Then the piece was fixed onto a glass slide, and an electrical lead was made by copper paste and copper wire insulated with silicone rubber.

2.3. Apparatus

The morphology of the as-prepared materials was investigated using scanning (SEM, Hitachi S4800) and transmission electron microscopy (TEM, JEM-2100). Electronic binding energies were measured using X-ray photoelectron spectroscopy (XPS) analysis which was performed on a PHI Quantum 2000 Scanning ESCA Microprobe with a monochromatized microfocused Al X-ray source. Energy dispersive X-ray spectroscopy (EDX) analysis was used to identify the elemental composition of the complex. The annealing process was conducted in Tube furnace (GR.TF60/11, ShanghaiGuier Machinery Co., Ltd). Electrochemical characterization was investigated using a CHI660B electrochemical workstation with a three-electrode system. A conventional three-electrode system was used with 3D NGA as the working electrode, saturated calomel electrode (SCE) as the reference electrode, and A Pt wire as the auxiliary electrode, respectively. Rotating ring-disk electrode (RDE) voltammetry was carried out on a speed controller (HP-1A, Jiangsu, China) and CHI 900C electrochemical workstation (Shanghai, China) using a modified GC ring-disk electrode (the GC electrode was polished with 1, 0.3 and 0.05 μm $\alpha\text{-Al}_2\text{O}_3$, sequentially. After ultrasonic concussion, the polished GC electrode was dried at room temperature. 5 μL 2 mg/mL of NGA suspension in Dimethylformamide obtained by sonication was dropped on the GC disk part for RDE, and then the electrode was subsequently dried at room temperature for 2 h.).

3. Results and discussion

3.1. characterization

In the experiments, the 3D-NGA could be obtained through the following procedures: (1) in-situ hydrothermal cross-linking and polymerization of the mixture to obtain the 3D hybrid N-containing precursor. During this process, DA turned to poly-DA between individual GO sheets and guided the graphene sheets to form a 3D structure (PDA-GO). (2) an annealing step at 800 $^{\circ}\text{C}$ under an Ar atmosphere was carried out to obtain an N doped character. As shown in Fig. 1a, the SEM image reveals that the wrinkled nanosheets were randomly cross-linked to form a network structure with rich pores. The 3D-NGA skeleton was dissected and investigated using TEM (Fig. 1b). We found that there is a typical corrugated structure with thin walls. XPS spectrums were

used to examine the 3D-NGA composition. The XPS peaks of the 3D-NGA at about 401, 284.8 and 532.7 eV corresponded to N1s, C1s and O1s (Fig. 1c). Compared with PDA-GO, 3D-NGA showed less content of oxygen, indicating the recovery of GO sp^2 structure. Moreover, as shown in the high-resolution N1s XPS spectra, e, before annealing, the N atoms of PDA-GO embodied the chemical bonding of PDA (primary amine, secondary amine, oxime) (Fig. 1e). Comparatively, after annealing, the N atoms of 3D-NGA were in three different valence states (pyridinic-N, graphitic-N and quaternary-N) in the graphene skeleton, suggesting that the N atoms were incorporated into the carbon-carbon bonds of graphene (Fig. 1d) [5]. Moreover, Fig. 1f showed the corresponding EDX spectrum of the peaks corresponding to C, O and N elements, then the atom content of the elements were figured out (C 87.64%, O 8.28%, N 4.07%).

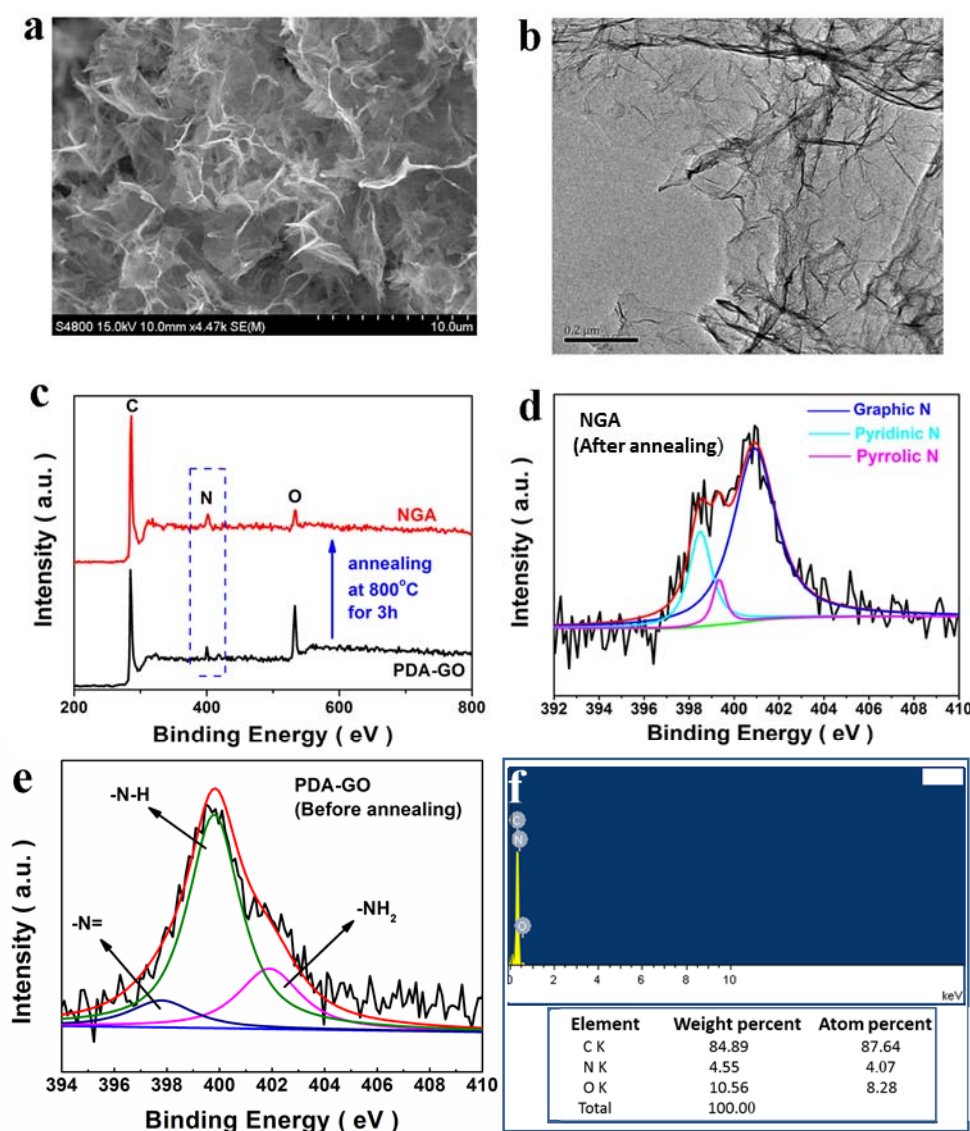


Fig. 1 (a) Typical SEM image and (b) TEM image of the NGA. (c) XPS spectrum of the NGA and PDA-GO, and

(d) (e) the corresponding high-resolution N1s peak. (f) the EDX spectrum of NGA.

The X-ray diffraction (XRD) patterns of NGA were also in comparison with PDA-GO (Fig. 2). PDA-GO synthesized using the hydrothermal approach usually has a broad diffraction peak at ca. 25.1° assigned to the (002) plane of stacked graphene sheets in the XRD pattern. The full width at half maximum of 3D NGA presents a relatively narrow peak. It is about 5° smaller than that of PDA-GO, indicating that the NGA has a relative higher graphitic structure.

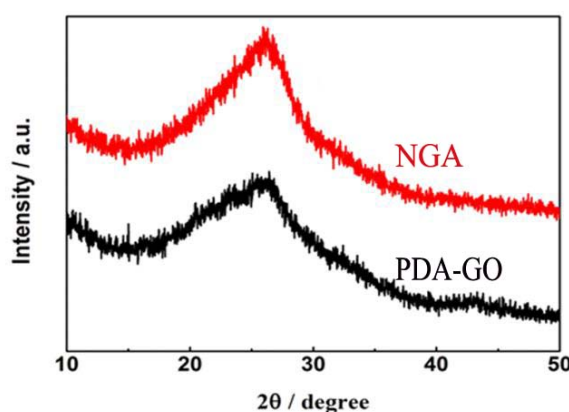


Fig. 2 XRD spectrum of the ultralight NGA and PDA-GO.

3.2. Electrocatalytic behaviors of the 3D-NGA

To examine the electrocatalytic characteristics of the 3D-NGA, cyclic voltammetric (CV) measurements were performed using the 3D-NGA electrode and non-doped 3D-GA electrode in the absence and presence of 5 mM H_2O_2 at a scan rate of 0.05 V s^{-1} (Fig. 3a). In the absence of H_2O_2 , it could be found that either 3D-GA electrode (curve 1) or the 3D-NGA electrode (curve 4) did not exhibit obvious redox peaks in the potential range from 0.2 to -0.8 V. In the presence of H_2O_2 , a remarkable catalytic current peak centered at -0.72 V could be observed for 3D-NGA electrode (curve 5), but the response of the 3D-GA electrode toward the reduction of H_2O_2 was very weak (curve 2). The above observations indicate that the 3D-NGA electrode is of highly electrocatalytic activity toward H_2O_2 reduction. The 3D structure provides a large accessible surface area which allows the easy diffusion of electrolyte into the inter region of the electrode. We also note that there was a relatively weaker cathodic current toward the electrocatalytic reduction of H_2O_2 (curve 3) using the 3D PDA-GO electrode. This result further confirmed that the intrinsic modification of graphene could effectively enhance its electron transfer ability.

It is reported that there was an intrinsic correlation between redox of H_2O_2 and ORR [20]. We further performed CV measurements using the bare 3D-GA electrode and the 3D-NGA electrode in O_2 and N_2 -saturated PBS (0.2 M, pH 7). As shown in Fig. 3b, no obvious reduction peak could be observed using both the 3D-GA electrode and the 3D-NGA electrode in N_2 saturated solution. In contrast, in the presence of O_2 , the 3D-GA electrode and the 3D-NGA electrode both exhibited a remarkable catalytic current peak about $-125 \mu\text{A}$ in intensity at -0.52 V and $-447 \mu\text{A}$ in intensity at -0.54 V , respectively, revealing that metal-free 3D-NGA electrode possessed electrocatalytic activity toward ORR. This result is quite consistent with the previous report [19].

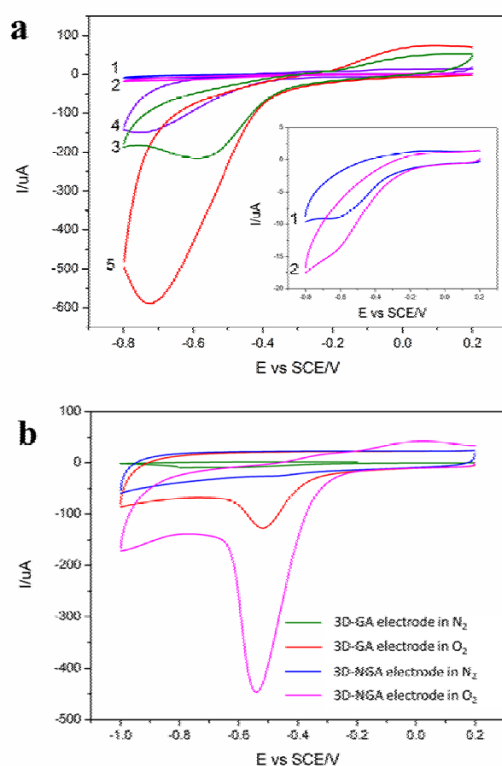


Fig. 3 (a) CVs of different electrodes in N_2 saturated 0.2 M PBS at pH 7 in the absence and presence of 5 mM H_2O_2 (curve 1, 4 represent 3D-GA electrode and 3D NGA electrode in the absence of H_2O_2 , respectively; curve 2, 5 represent 3D-GA electrode and 3D NGA electrode in the presence of H_2O_2 , respectively; curve 3 represent 3D PDA-GO electrode in the presence of H_2O_2). (b) CVs of a 3D-GA and 3D-NGA in the presence of N_2 or O_2 .

To further assess the ORR and OER catalytic activity of the synthesized nanocomposites, electrochemical tests were carried out using a rotational ring-disk electrode (RDE) in alkaline solution. Figure 3 showed the typical sweeping voltammograms using the RDE at an electrode rotating speed of 1600 rpm and a potential scanning rate of 5 mV s^{-1} . As shown in the linear sweep voltammograms (LSVs), 3D-NGA electrode afforded an onset potential of 0.89 V versus

reversible hydrogen electrode (RHE), and a half-wave potential ($E_{1/2}$) of 0.66 V in the ORR region (Fig. 4a), which is comparable with those of Pt/C (onset potential = 0.94 V, $E_{1/2}$ = 0.80 V). In OER region, 3D-NGA afforded a sharp onset potential at 1.46 V, which is lower than that of Pt/C at 1.52 V (Fig. 4b). The OER overpotentials at a current density of 10 mA cm⁻² is also lower than that of Pt/C. This catalytic performance both in ORR and OER indicates that 3D-NGA electrode can be counted as a promising low-cost bifunctional electrocatalyst, which therefore confirm its ability to catalyze H₂O₂ into H₂O.

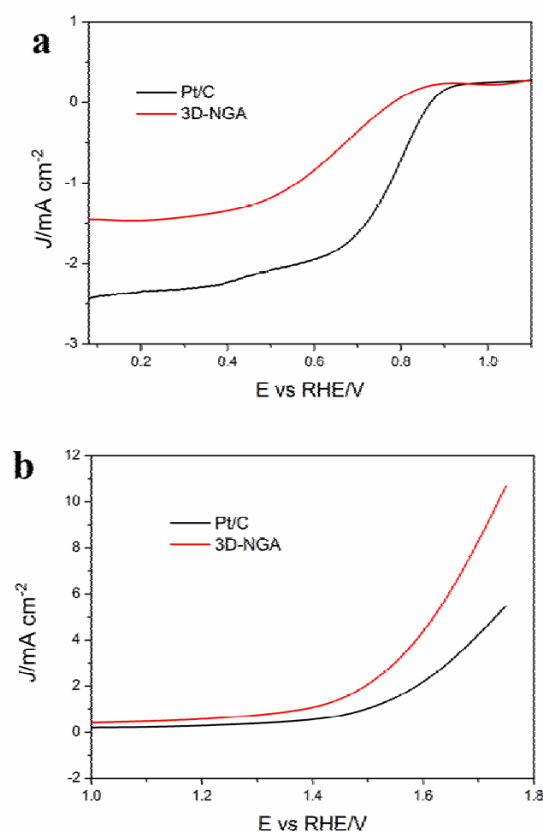


Fig. 4 (a) ORR activity of 3D-NGA and Pt/C measured using RDEs in O₂-saturated 0.1 M KOH. (b) OER activity of 3D-NGA and Pt/C using RDEs in O₂-saturated 0.1 M KOH.

3.3. Sensing of H₂O₂

Amperometric *i-t* technique is an effective way to examine the sensing property of the fabricated electrochemical sensor. In order to improve the 3D-NGA electrode performance towards non-enzymatic H₂O₂ sensing, we investigated the effect of the applied potential on the amperometric response on the 3D-NGA electrode towards the H₂O₂ reduction by varying the potential between -0.7 V and -0.3 V. As shown in Fig. 5ab, the optimal applied potential at -0.5 V

was selected. The pH effect of PBS on the amperometric response was also examined. The profile of Fig. 5cd, showed that the current reached a maximum value at pH 7. Therefore, pH 7 PBS was selected in the further studies.

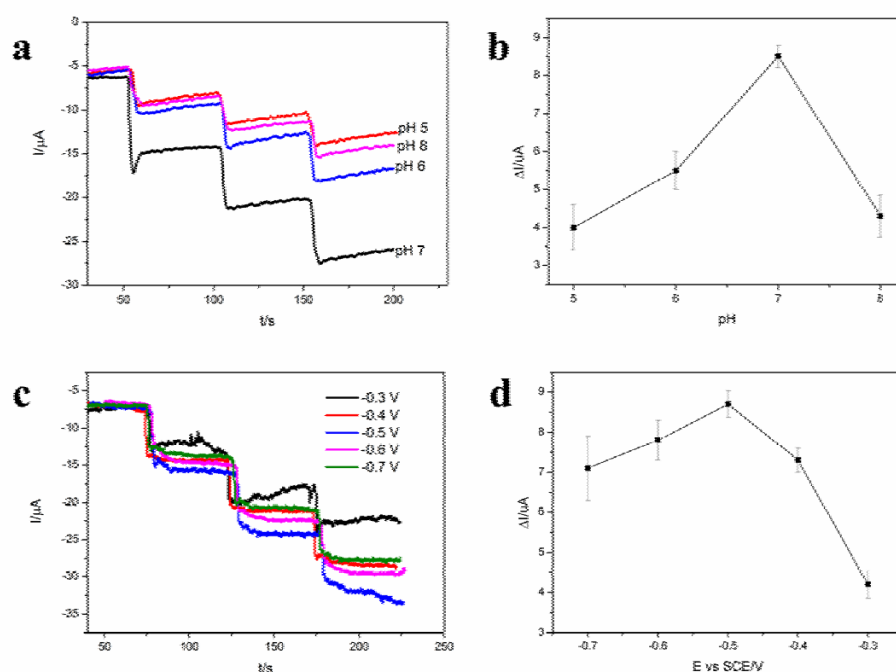


Fig. 5 The effect of solution pH (a) (b) and applied potential (c) (d) to the amperometric response of 1 mM H_2O_2 on the 3D-NGA electrode.

Fig. 6 shows a typical current-time plot on the successive step change of H_2O_2 concentrations using the 3D-NGA electrode in stirring 0.2 M PBS buffer (pH 7, N_2 -saturated) at the applied potential of -0.5 V. When an aliquot of H_2O_2 was dropped into the stirring solution, the reduction current increased steeply to reach a stable value within 3 s, indicating its fast amperometric response behavior. The inset in Fig. 6 shows the linear relationship between the catalytic currents and the concentrations of H_2O_2 . The catalytic current increased with the increase of H_2O_2 from 0.2 to 35 mM ($R^2 = 0.9950$) with a detection limit ($S/N = 3$) of 0.05 mM, which are comparable to or even better than those of the other metal-free or enzymebased H_2O_2 biosensors [26-30].

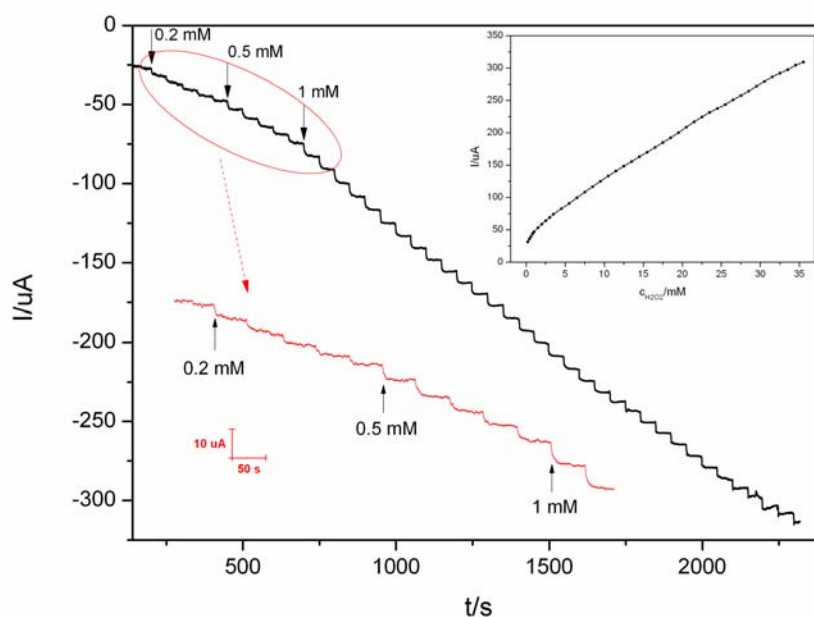


Fig. 6 Amperometric response of the 3D-NGA electrode to successive injection of H_2O_2 into the stirred PBS.

The inset is the calibration curve.

To investigate the selectivity for H_2O_2 sensing, the amperometric responses of ascorbic acid (AA), dopamine (DA), uric acid (UA) and glucose are studied using the 3D-NGA electrode. In the experiments, The working potential was kept at -0.5 V , the concentrations of H_2O_2 and the co-existing interfering species were all 1 mM . As shown in Fig. 7, no obvious current change could be observed upon the addition of the each interfering specie. Our fabricating sensor demonstrates high selectivity and reliable anti-interference property by comparing the amperometric responses of other relevant electroactive species. The stability and reproducibility of the sensor are also essential for the monitoring of H_2O_2 . The sensitivity could still keep above 95% of its initial sensitivity even if the 3D-NGA electrode was exposed in air for two weeks, revealing the satisfactory stability of the as-prepared nonenzymatic H_2O_2 sensor. The reproducibility of the approach was also investigated and the relative standard deviation (RSD) for H_2O_2 sensing was less than 6.3% for 10 measurements using a same electrode, indicating its good selectivity and stability for H_2O_2 sensing. Therefore, the as-prepared 3D-NGA electrode is an excellent candidate for the fabrication of stable and specific amperometric sensor for the non-enzymatic detection of H_2O_2 .

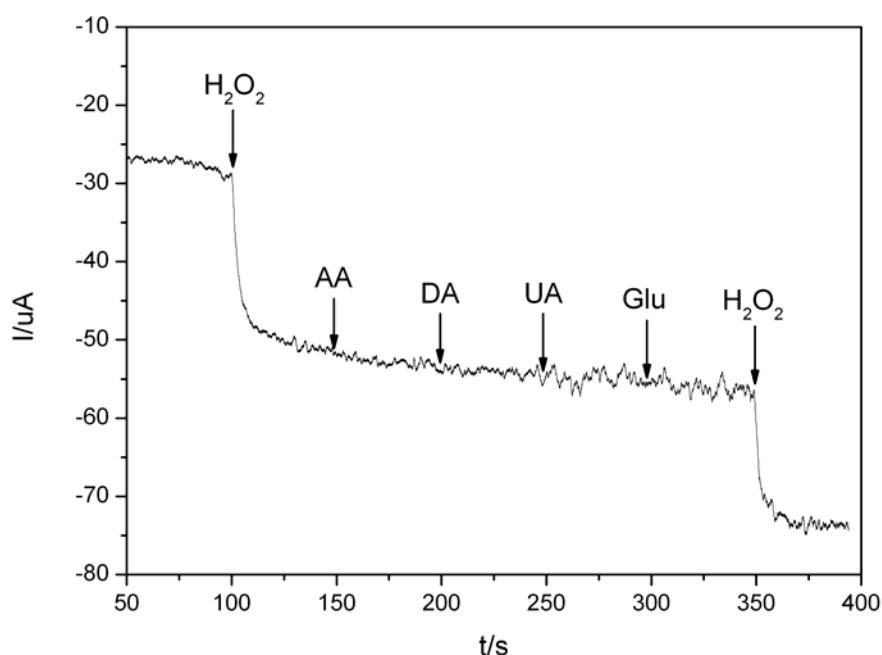


Fig. 7 Amperometric response of the 3D-NGA electrode exposed to H₂O₂, AA, DA, UA and glucose.

4. Conclusions

In this study, a novel 3D N-doped graphene aerogel was prepared and used as a sensing material for non-enzymatic electrochemical reduction of H₂O₂. Owing to the 3D open-pore structure and nitrogen doping property, the 3D-NGA exhibited excellent electrocatalytic activity and sensing ability towards H₂O₂ in terms of detection range, stability and response time. Due to the ease of synthesis and electrode fabrication, good specificity and stability, 3D-NGA becomes promising microstructures for electrochemical biosensor devices of H₂O₂. In addition, the 3D-NGA can also be used as an ORR and OER electrocatalyst without any metal. The results demonstrated that the good nanostructured material, 3D-NGA, is of the potential applications in biosensor and low-cost fuel cells construction.

Acknowledgements

This research work was financially supported by the National Nature Scientific Foundation of China (No. 21175112 and 21375112), Program for Changjiang Scholars and Innovative Research Team in University (PCSIRT, No. IRT13036) and NFFTBS (No. J1310024). They were gratefully

acknowledged.

References

- [1] M.J. Allen, V.C. Tung, R.B. Kaner, Honeycomb Carbon: A Review of Graphene, *Chem. Rev.* 110 (2009) 132-145.
- [2] R.H. Baughman, A.A. Zakhidov, W.A. de Heer, Carbon Nanotubes-the Route Toward Applications, *Science* 297 (2002) 787-792.
- [3] M.C. Rodríguez, J. Sandoval, L. Galicia, S. Gutiérrez, G.A. Rivas, Highly selective determination of uric acid in the presence of ascorbic acid at glassy carbon electrodes modified with carbon nanotubes dispersed in polylysine, *Sens. Actuators B* 134 (2008) 559-565.
- [4] Z. H. Sheng, X. Q. Zheng, J. Y. Xu, W. J. Bao, F. B. Wang, X. H. Xia, Electrochemical sensor based on nitrogen doped graphene: Simultaneous determination of ascorbic acid, dopamine and uric acid, *Biosens. Bioelectron.* 34 (2012) 125-131.
- [5] X. Song, L. Lin, M. Rong, Y. Wang, Z. Xie, X. Chen, Mussel-inspired, ultralight, multifunctional 3D nitrogen-doped graphene aerogel, *Carbon* 80 (2014) 174-182.
- [6] F.H. Wu, G.C. Zhao, X.W. Wei, Electrocatalytic oxidation of nitric oxide at multi-walled carbon nanotubes modified electrode, *Electrochem. Commun.* 4 (2002) 690-694.
- [7] X. Xu, S. Jiang, Z. Hu, S. Liu, Nitrogen-Doped Carbon Nanotubes: High Electrocatalytic Activity toward the Oxidation of Hydrogen Peroxide and Its Application for Biosensing, *ACS Nano* 4 (2010) 4292-4298.
- [8] C.C. Hwang, G. Ruan, L. Wang, H. Zheng, E.L.G. Samuel, C. Xiang, W. Lu, W. Kasper, K. Huang, Z. Peng, Z. Schaefer, A.T. Kan, A.A. Martí, M.S. Wong, M.B. Tomson, J.M. Tour, Carbon-Based Nanoreporters Designed for Subsurface Hydrogen Sulfide Detection, *ACS Appl. Mater. Interfaces* 6 (2014) 7652-7658.
- [9] A.H. Pourasl, M.T. Ahmadi, M. Rahmani, H.C. Chin, C.S. Lim, R. Ismail, M.L.P. Tan, Analytical modeling of glucose biosensors based on carbon nanotubes, *Nanoscale Res. Lett.* 9 (2014) 33.
- [10] J. Wang, S.B. Hocevar, B. Ogorevc, Carbon nanotube-modified glassy carbon electrode for adsorptive stripping voltammetric detection of ultratrace levels of 2,4,6-trinitrotoluene, *Electrochem. Commun.*, 6 (2004) 176-179.
- [11] L. Chen, X. Wang, X. Zhang, H. Zhang, 3D porous and redox-active prussian blue-in-graphene aerogels for highly efficient electrochemical detection of H₂O₂, *J. Mater. Chem.* 22 (2012) 22090-22096.
- [12] C. Zhou, J. Kong, E. Yenilmez, H. Dai, Modulated Chemical Doping of Individual Carbon Nanotubes, *Science* 290 (2000) 1552-1555.
- [13] F. Cheng, J. Chen, Metal-air batteries: from oxygen reduction electrochemistry to cathode catalysts, *Chem. Soc. Rev.* 41 (2012) 2172-2192.
- [14] X. Han, F. Cheng, T. Zhang, J. Yang, Y. Hu, J. Chen, Hydrogenated Uniform Pt Clusters Supported on Porous CaMnO₃ as a Bifunctional Electrocatalyst for Enhanced Oxygen Reduction and Evolution, *Adv. Mater.* 26 (2014) 2047-2051.
- [15] J.I. Jung, H.Y. Jeong, J.S. Lee, M.G. Kim, J. Cho, A Bifunctional Perovskite Catalyst for Oxygen Reduction and Evolution, *Angew. Chem. Int. Ed.* 53 (2014) 4582-4586.
- [16] Y. Zhao, R. Nakamura, K. Kamiya, S. Nakanishi, K. Hashimoto, Nitrogen-doped carbon nanomaterials as non-metal electrocatalysts for water oxidation, *Nat. Commun.* 4 (2013) 2390.

- [17] D.W. Wang, D. Su, Heterogeneous nanocarbon materials for oxygen reduction reaction, *Energ. Environ. Sci.*, 7 (2014) 576-591.
- [18] T.Y. Ma, S. Dai, M. Jaroniec, S.Z. Qiao, Graphitic Carbon Nitride Nanosheet-Carbon Nanotube Three-Dimensional Porous Composites as High-Performance Oxygen Evolution Electrocatalysts, *Angew. Chem. Int. Ed.* 53 (2014) 7281-7285.
- [19] L. Qu, Y. Liu, J.B. Baek, L. Dai, Nitrogen-Doped Graphene as Efficient Metal-Free Electrocatalyst for Oxygen Reduction in Fuel Cells, *ACS Nano* 4 (2010) 1321-1326.
- [20] Y. Tang, B.L. Allen, D.R. Kauffman, A. Star, Electrocatalytic Activity of Nitrogen-Doped Carbon Nanotube Cups, *J. Am. Chem. Soc.* 131 (2009) 13200-13201.
- [21] R.A. Sidik, A.B. Anderson, N.P. Subramanian, S.P. Kumaraguru, B.N. Popov, O₂ Reduction on Graphite and Nitrogen-Doped Graphite: Experiment and Theory, *J. Phys. Chem. B* 110 (2006) 1787-1793.
- [22] Y. Wang, Y. Shao, D.W. Matson, J. Li, Y. Lin, Nitrogen-Doped Graphene and Its Application in Electrochemical Biosensing, *ACS Nano* 4 (2010) 1790-1798.
- [23] F. Xu, Y. Sun, Y. Zhang, Y. Shi, Z. Wen, Z. Li, Graphene-Pt nanocomposite for nonenzymatic detection of hydrogen peroxide with enhanced sensitivity, *Electrochem. Commun.* 13 (2011) 1131-1134.
- [24] X. Chen, Z. Cai, Z. Huang, M. Oyama, Y. Jiang, X. Chen, Ultrafine palladium nanoparticles grown on graphene nanosheets for enhanced electrochemical sensing of hydrogen peroxide, *Electrochim. Acta* 97 (2013) 398-403.
- [25] H.A. Becerril, J. Mao, Z. Liu, R.M. Stoltenberg, Z. Bao, Y. Chen, Evaluation of Solution-Processed Reduced Graphene Oxide Films as Transparent Conductors, *ACS Nano* 2 (2008) 463-470.
- [26] V. Penmatsa, T. Kim, M. Beidaghi, H. Kavarada, L. Gu, Z. Wang, C. Wang, Three-dimensional graphene nanosheet encrusted carbon micropillar arrays for electrochemical sensing, *Nanoscale* 4 (2012) 3673-3678.
- [27] J. Tian, Q. Liu, C. Ge, Z. Xing, A.M. Asiri, A.O. Al-Youbi, X. Sun, Ultrathin graphitic carbon nitride nanosheets: a low-cost, green, and highly efficient electrocatalyst toward the reduction of hydrogen peroxide and its glucose biosensing application, *Nanoscale* 5 (2013) 8921-8924.
- [28] Z. Cao, X. Jiang, Q. Xie, S. Yao, A third-generation hydrogen peroxide biosensor based on horseradish peroxidase immobilized in a tetrathiafulvalene-tetracyanoquinodimethane/multiwalled carbon nanotubes film, *Biosens. Bioelectron.* 24 (2008) 222-227.
- [29] C. Guo, Y. Song, H. Wei, P. Li, L. Wang, L. Sun, Y. Sun, Z. Li, Room temperature ionic liquid doped DNA network immobilized horseradish peroxidase biosensor for amperometric determination of hydrogen peroxide, *Anal. Bioanal. Chem.*, 389 (2007) 527-532.
- [30] J. Tian, H. Li, W. Lu, Y. Luo, L. Wang, X. Sun, Preparation of Ag nanoparticle-decorated poly(m-phenylenediamine) microparticles and their application for hydrogen peroxide detection, *Analyst* 136 (2011) 1806-1809.

Highlights

- A novel 3D nitrogen (N) doped graphene aerogel (GA) was used as a direct electrode for

electrochemical sensing.

- Owing to the 3D open-pore structure and nitrogen doping property, the sensor showed excellent electrocatalytic activity towards H_2O_2 reduction.
- The proposed non-enzymatic sensor was applied for H_2O_2 detection.

Author Biographies

Zhixiong Cai is currently a PhD. student majoring in analytical chemistry in College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, China. He is mainly engaged in the research of electroanalytical chemistry.

Xinhong Song is currently a PhD. student majoring in analytical chemistry in College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, China. Her interest lies in carbon nanostructures for sensor fabrication and catalysis.

Yiying Chen is currently a master student majoring in analytical chemistry in College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, China. She is mainly engaged in the research of the preparation of carbon nanostructures.

Yiru Wang is an Associate Professor of Xiamen University in China. She received PhD degree in 1999. Her research interests cover the sample pretreatment techniques for environmental analysis, food analysis and pharmaceutical analysis.

Xi Chen is a Full Professor in the Department of Chemistry at Xiamen University, China. He received his PhD (1996) degree in Analytical Chemistry from Kyoto Institute of Technology, Japan. Dr Chen's research interests cover electrochemiluminescence, bio-chemical sensors and solid-phase microextraction. He has published over 100 scientific research articles in archival scientific journals.